

Md Badrul Hasan

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RESEARCH SUMMARY

My research advances scientific machine learning methods for multi-physics simulation, combining physics-constrained neural networks and data-driven SGS closure modeling to simulate spatiotemporal atmospheric boundary-layer dynamics. I develop invariance-embedded ML surrogates integrated into large-scale CFD solvers (e.g., WRF), leveraging HPC to enable high-fidelity hurricane boundary-layer simulations at scale.

EDUCATION

University of Maryland, Baltimore County (UMBC) 2022 - June, 2026

Ph.D. in Mechanical Engineering

Advisor: Dr. Meilin Yu

University of Maryland, Baltimore County (UMBC) 2019 - 2022

M.S. in Mechanical Engineering

Advisors: Dr. Meilin Yu, Dr. Stephen Guimond

Bangladesh University of Engineering and Technology (BUET) 2013 - 2017

B.Sc. in Mechanical Engineering

WORK EXPERIENCE

Harold Vance Department of Petroleum Engineering,

Texas A&M University

October, 2026 - Present

Postdoctoral Researcher

- Accelerating the Fortran-based XBeach coastal storm-impact solver on GPUs using nvFortran and OpenACC, on an ONR-funded project with Dr. Wencheng Jin (PI) and Deltares as subawardee.
- Established a profiling-first offload strategy from CPU profiling and MPI strong-scaling campaigns on a 72-hour Hurricane Sandy hindcast (84,537 cells), identifying the wave-action solver as the dominant bottleneck (53–86% of runtime across directional resolutions) and measuring $51\times$ strong scaling at 143 MPI ranks.
- Verified solver correctness across compilers: seeded gfortran and nvFortran builds driven by byte-identical boundary conditions agree on final bed level to bitwise round-off.
- Developing JAX-based machine-learning closures for wave hydrodynamics and a neural emulator for morphodynamics to complement GPU offloading.

Computational Mechanics Laboratory, UMBC

January, 2022 - August, 2026

Graduate Research Assistant

- Conducted *a posteriori* and *a priori* evaluations of invariance-embedded machine learning SGS closures for rotating hurricane boundary layers, integrating ML-based spatio-temporal stress and eddy-viscosity surrogates into WRF. (**UMBC 2025 COEIT Interdisciplinary Proposal Award**)
- Developed machine-learning-based models of backscatter-admitting sub-grid-scale (SGS) processes to improve hurricane boundary layer simulations.
- Used advanced methods for detecting stealthy, long-term cyber-attacks on wind energy assets using physics-informed neural networks. (**UMBC 2024 Cybersecurity Leadership Exploratory Grant**)

- Contributed to the development of next-generation naval design tools using CFDship-Iowa, in collaboration with industrial and academic partners.

Joint Center for Earth Systems Technology (JCET), UMBC

January, 2020 - June, 2022

Graduate Research Assistant

- Investigated numerical dissipation across weather prediction models (WRF and NUMA) with Dr. Stephen Guimond; supported by NSF grant AGS-2121366.
- Analyzed remote-sensing radar observations from the Imaging Wind and Rain Airborne Profiler (IWRAP) with Dr. Stephen Guimond for the NOAA/AOML/HRD Hurricane Field Program.

University of Maryland, Baltimore County (UMBC)

January, 2019 - May, 2024

Graduate Teaching Assistant

- Conducted the lab demonstrations and grading on ENME-432L, Fluids/Energy Lab with Dr. Meilin Yu.
- Assisted Dr. James Baughan on teaching and grading on the ENME-423, HVAC Design.

PUBLICATIONS

Journals

- **Hasan, M. B.**, Guimond, S. R., Yu, M., Reddy, S., & Giraldo, F. X. (2022). The Effects of Numerical Dissipation on Hurricane Rapid Intensification with Observational Heating. *Journal of Advances in Modeling Earth Systems*, 14, e2021MS002897. [DOI]
- **Hasan, M. B.**, Yu, M., & Oates, T. (2026), Invariance-embedded Machine Learning Sub-grid-scale Stress Models for Meso-scale Hurricane Boundary Layer Flow Simulation I: Model Development and *a priori* Studies. *International Journal For Numerical Methods In Fluids*. [DOI]
- **Hasan, M. B.**, Yu, M., & Oates, T. Invariance-Embedded Machine Learning Sub-Grid-Scale Stress Models for Meso-Scale Hurricane Boundary Layer Flow Simulation: A Posteriori Integration and Evaluation. (In Preparation, 2026)
- **Hasan, M. B.**, Kalwani, S., Chen, Z. & Yu, M. Detecting Stealthy Cyber Attacks on Wind Energy Assets with ODE Physics-Informed Learning and Adversarial Hardening. (In Preparation, 2026)

Conference Proceedings

- **Hasan, M. B.**, Yu, M., Xiao, H. (2023). Sub-grid Scale Modeling of Meso-scale Hurricane Boundary Layer Flows using Machine Learning. In *AIAA SciTech 2023 Forum*, p. 2487. [DOI]
Presented at AIAA SciTech 2023, National Harbor, MD.
- **Hasan, M. B.**, Yu, M., Oates, T. (2025). Comparison of Several Machine-Learning-Enhanced Sub-grid Scale Stress Models for Meso-scale Hurricane Boundary Layer Flow Simulation. In *AIAA SciTech 2025 Forum*, p. 2212. [DOI]
Presented at AIAA SciTech 2025, Orlando, FL.
- **Hasan, M. B.**, Yu, M., Oates, T. (2025). Evaluating Machine Learning-Enhanced Sub-Grid Scale Stress Models With Invariance Embedding for Meso-Scale Hurricane Boundary Layer Flows. In *Proceedings of the ASME 2025 Fluids Engineering Division Summer Meeting, V001T01A001.*, [DOI]
Presented at ASME FEDSM 2025, Philadelphia, PA.
- Kalwani, S., **Hasan, M. B.**, Chen, Z., Yu, M., Physics-Informed Machine Learning for Detecting Stealthy Long-Term Cyber Attacks on Wind Energy Systems, 2025 IEEE International Conference on Big Data (BigData), Macau, China, 2025, pp. 6587-6596, [DOI]

- **Hasan, M. B.**, Yu, M., Oates, T. (2026). Semi-A Priori Evaluation of Backscatter-Admitting Machine Learning Sub-Grid Scale Models in WRF Hurricane Simulations. In *AIAA Aviation 2026 Forum*. [DOI] Presented at AIAA AVIATION 2026, San Diego, CA.
- **Hasan, M. B.**, Yu, M., Oates, T. (2026). A Baseline A Posteriori Evaluation of Machine-Learning-Predicted Eddy Viscosity Fields in Mesoscale Hurricane Boundary Layer Simulations. In *ASME Fluids Engineering Division Summer Meeting (FEDSM)*, 2026. Presented at ASME FEDSM 2026, Bellevue, WA.

CONFERENCE PRESENTATIONS & SEMINARS

- Machine Learning-Enhanced Turbulence Modeling for Hurricane Boundary Layer Simulations, *Poster Session, COEIT Research Day*, UMBC, Baltimore, MD, May 2026. [**Poster, Doctoral Student Award**]
- Schroeder, R., **Hasan, M. B.**, Yu, M. Data-Driven Modeling of Dynamic Stall in Vertical-Axis Wind Turbines, *Poster Session, COEIT Research Day*, UMBC, Baltimore, MD, May 2026. [**Poster, co-author, Undergraduate Student Award**]
- Physics-Informed Machine Learning for Turbulence, Hurricanes, and Wind-Energy Cybersecurity, *Mechanical Engineering Graduate Seminar*, University of Maryland, Baltimore County (UMBC), February 2026. [**Oral**]
- Invariance-Embedded Machine Learning Sub-Grid-Scale Stress Models for Meso-Scale Hurricane Boundary Layer Simulations, *2025 Research Symposium on Environmental and Applied Fluid Dynamics*, The George Washington University, Washington, DC, May 2025. [**Oral**]
- Kalwani, S., **Hasan, M. B.**, Chen, Z., Yu, M. Stealthy Long-Term Cyber Attacks Detection on Wind Turbines Using Physics-Informed Neural Networks, *Poster Session, COEIT Research Day*, UMBC, Baltimore, MD, April 2025. [**Poster, co-author, Honorable Mention Master Student Award**]
- Assessment of Invariance-Embedded Machine Learning Models for Sub-Grid Scale Stress in Meso-Scale Hurricane Boundary Layer Flows, *COEIT Research Day*, UMBC, Baltimore, MD, April 2025. [**Oral**]
- Sub-grid Scale Modeling of Meso-Scale Hurricane Boundary Layer Flows using Machine Learning, *COEIT Research Day*, UMBC, Baltimore, MD, April 2024. [**Oral**]
- The Effects of Numerical Dissipation on Hurricane Rapid Intensification with Observational Heating, *AGU Fall Meeting*, New Orleans, LA, Dec 2021. [**Poster**]
- The Effects of Numerical Dissipation on Simulating Hurricane Intensification in a Realistic Regime, *AGU Fall Meeting*, San Francisco, CA, Dec 2020. [**Poster**]
- The Effects of Numerical Dissipation on Simulating Hurricane Intensification in a Realistic Regime, *Seminar Series*, Department of Mechanical Engineering, UMBC, Baltimore, MD, Nov 2020. [**Oral**]

AWARDS & SCHOLARSHIPS

2026–2027 ASME Fluids Engineering Division Ghia Scholar

Urmila and Karman Ghia FED Graduate Student Scholar (GSS) Award – ASME-sponsored recognition for international graduate students in the U.S. studying computational fluid dynamics; assigned to the Computational Fluid Dynamics Technical Committee (CFDTC), ASME FEDSM 2026, Bellevue, WA.

2026 COEIT Research Day Doctoral Student Award (Poster)

Awarded for the poster “Machine Learning-Enhanced Turbulence Modeling for Hurricane Boundary Layer Simulations,” COEIT Research Day, UMBC, May 2026.

2025 AIAA Professor Kirti “Karman” Ghia Memorial Award

Inaugural recipient for best student paper, Comparison of Several Neural Network-Enhanced Sub-grid Scale Stress Models for Meso-scale Hurricane Boundary Layer Flow Simulation, awarded by the Fluid Dynamics Technical Committee (FDTC) at AIAA SciTech 2025.

Graduate Student Association (GSA) Professional Development Grant, UMBC

Recipient in **May, 2026**; **May, 2025** and **December, 2024**;

Provided support for professional development and thesis-related research expenses.

University Technical Scholarship (2013–2017)

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

GRANT & PROPOSAL EXPERIENCE

NSF Research Proposal (Submitted, 2026)

Led technical development of the scientific methodology and contributed to writing for a multi-year NSF proposal integrating physics-informed machine learning with wind energy cyber-physical systems. Included solver verification and validation components and multi-physics modeling of coupled fluid-electrical systems.

TECHNICAL STRENGTHS

Turbulence & SGS Modeling	SGS closures, eddy-viscosity models, rotating boundary layers
CFD / PDE Models	WRF, NUMA, ANSYS Fluent, COMSOL, OpenFAST, CFDship-Iowa, data-integrated simulation workflows
Scientific Machine Learning	PINNs, neural operators, physics-constrained surrogates, spatio-temporal modeling
Deep Learning	PyTorch, CUDA
High-Performance Computing	Slurm, OpenMPI, GPU clusters
Programming	Python (PyTorch, NumPy, xarray), MATLAB, Fortran, Bash

WORKSHOPS & TRAINING

Structure-Preserving Scientific Computing and Machine Learning: Summer School & Hackathon,
University of Washington, Seattle June, 2025

Participant

- Selected as one of **40 graduate students** from across the U.S. and Canada; program supported in part by **NSF** and **PIMS**.
- Hackathon **Project D: Neural ODEs Exploring Time Integration Methods and Training Strategies**; relevant to *weather forecasting* and *nonlinear dynamical systems*.
- Activities included lectures, hands-on computational labs, and collaborative mini-projects at the intersection of scientific computing and machine learning.

LEADERSHIP & INVOLVEMENT

Bangladesh Student Association, UMBC

September, 2019 -August, 2020

Treasurer

- Managed funds and coordinated events for Bangladeshi graduate students at UMBC, fostering community engagement.

MENTORING EXPERIENCE

Undergraduate Research Mentor

2025 - 2026

Computational Mechanics Laboratory, UMBC

- Mentoring undergraduate researcher Robby Schroeder on a data-driven feedforward neural network framework for predicting dynamic stall in vertical-axis wind turbines (VAWTs), funded by the UMBC Technology Catalyst Fund.
- Guided model architecture design, training strategy, and validation against CFD reference data; lift and drag prediction errors reduced from 28% to 7.2% and 40% to 4.2%, respectively.
- Supported poster development for COEIT Research Day 2026, which received the **Undergraduate Student Award**.

PEER REVIEW SERVICE

- **Physics of Fluids** (AIP Publishing) - computational fluid dynamics
- **AIAA Aviation Forum 2026** - data-driven methods for CFD modeling
- **ASME FEDSM 2027** - Computational Fluid Dynamics Technical Committee (CFDTC)

MEDIA COVERAGE & ONLINE FEATURES

UMBC News Feature

- **Modeling Hurricanes with Machine Learning**
Research featured in UMBC News highlighting ML-based hurricane modeling. (January 2025)
- **Mechanical engineering Ph.D. alumnus recognized with ASME scholar award**
Recognition by UMBC COEIT for the ASME scholar award. (August 2026)